

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

2. **Answer:** D

Options: A) 0.64; B) 0.72; C) 0.81; D) 0.95

Note: Correct option: D - 0.95

3. **Answer:** A

Options: A) $(x - 2)(x + 2)(x - 1)(x + 1)$; B) $(x+1)^4$; C) $(x-1)(x+1)^3$; D) $(x-1)^3(x+1)$

Note: Correct option: A - $(x - 2)(x + 2)(x - 1)(x + 1)$

4. **Answer:** C

Options: A) $g = g^{-1}$ for every g in G ; B) $g = g^2$ for every g in G ; C) $(g \circ h)^2 = g^2 \circ h^2$ for every g, h in G ; D) G is of finite order

Note: Correct option: C - $(g \circ h)^2 = g^2 \circ h^2$ for every g, h in G

5. **Answer:** A

Options: A) Sunday; B) Monday; C) Tuesday; D) Wednesday

Note: Correct option: A - Sunday

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'False, True'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'between 0.02 and 0.16'.

Long Form Response

11. **Answer:** Let a be the number of colonies on Earth-like planets and b be the number on Mars-like planets. We therefore seek nonnegative integers a and b such that $2a+b = 12$. From this equation, we see that b can be at most 6 while a can at most be 5. In addition, b must be even, so the only possibilities are $b = 6, 4, 2$. Thus, there are 3 possible colonization options: $a = 3, b = 6$; $a = 4, b = 4$; $a = 5, b = 2$. In the first option, we take all 6 Mars-like planets, and can choose the Earth-like planets in $53 = 10$ ways. This gives us 10 possibilities. In the second option, we can choose any 4 of the 5 like Earth and any 4 of the 6 like Mars. This is $5464 = 75$ possibilities. In the third option, all Earth-like planets must be occupied while only 2 of those like Mars must be occupied. This is $5562 = 15$ possibilities. In all, there are $10 + 75 + 15 = 100$ planets. *Note:* Students should show all steps in their mathematical solution.

12. **Answer:** The exterior angles of a triangle sum to 360° , so the average of their measures is $360^\circ/3 = 120^\circ$. *Note:* Students should show all steps in their mathematical solution.